Avinash Kumar Singh

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Education

Vellore Institute of Technology Sep 2023 – May 2027
B.Tech in Computer Science Engineering (Cyber Security and Digital Forensics) *CGPA:* ***8.14***

Experience

Cybersecurity Analyst Intern — *Zorvyn FinTech Pvt. Ltd.* *Apr 2026 – Sep 2026*

- Assisted in monitoring and triaging endpoint and network security alerts using **Splunk SIEM**, analyzing event logs and suspicious activity to support investigations and escalation.
- Prepared **10+ vulnerability assessment and security incident reports**, documenting **CVSS**-based severity, proof-of-concept evidence, IoCs where applicable, potential impact, and remediation recommendations.
- Collaborated with **5+ developers** to prioritize security findings, track patch management, and validate fixes, contributing to a reduction in average vulnerability remediation time from **10 days to 7 days**.

Projects

Rakshak: SOC Alert Triage & Threat Analysis Tool | *Python, FastAPI, MySQL, MITRE ATT&CK*
github.com/avinash09-cmd/Rakshak

- Developed a Python-based SOC alert triage tool using **FastAPI** and **MySQL** to receive, organize, and track security alerts.
- Implemented rule-based alert classification using predefined conditions to categorize suspicious activity and assign initial severity levels.
- Mapped relevant alert patterns to **MITRE ATT&CK** techniques to support threat identification and investigation.
- Created API endpoints to store alert details, classification results, and investigation notes, supporting security event review and escalation.
- Planned future enhancement to integrate an **LLM** for automated alert summarization, contextual threat analysis, and investigation support to improve SOC alert triage efficiency.

Checkmate: Secure Task Management REST API | *Python, FastAPI, MySQL, JWT, Docker, GitHub Actions*
github.com/avinash09-cmd/-Checkmate-OWASP-Protected-Secure-REST-API

- Developed a task-management web application using **FastAPI** and **MySQL**, implementing **JWT** authentication and **role-based access control (RBAC)** to manage user permissions.
- Applied **STRIDE** threat modeling to identify potential threats, including unauthorized access, data tampering, and information disclosure, and guide security control selection.
- Implemented task-level authorization, **Argon2** password hashing, **Pydantic** input validation, parameterized SQL queries, and rate limiting to mitigate common web application security risks.
- Integrated automated testing and security checks using **pytest**, **Bandit**, **Semgrep**, **pip-audit**, and **OWASP ZAP** to identify potential vulnerabilities and validate fixes.

Technical Skills

- **Programming Languages:** Java, Python, SQL, HTML, CSS, JavaScript
- **Cybersecurity:** SOC Alert Triage, Log Analysis, Vulnerability Assessment, CVSS, MITRE ATT&CK, STRIDE & DREAD Threat Modeling, OWASP Top 10
- **Security Tools:** Splunk SIEM, Nmap, Wireshark, OWASP ZAP, Bandit, Semgrep, pip-audit
- **Frameworks & Development:** FastAPI, MySQL, JWT, RBAC, Argon2, Pydantic, pytest, Docker, Git, GitHub Actions
- **Networking:** TCP/IP, DNS, HTTP/HTTPS, Ports and Protocols, Firewalls, Network Traffic Analysis
- **Relevant Coursework:** Computer Networks, Operating Systems, DBMS, Digital Forensics, Cyber Threat Analysis, Software Vulnerability Testing

Achievements & Certifications

- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate** (2025)
- **Google IT Support Professional Certificate** (2026)
- **Certified Ethical Hacker (CEH)** — In progress
- **Top-5 Finalist** (200+ teams), Anti-Piracy Hackathon — WAVES Summit, MSME, Govt. of Maharashtra
- **Vice President, Metaversity Club** — Led 23 members, organized 5 tech talks, grew participation 40%
- **National-Level Multi-sport Athlete** — Represented university in basketball and swimming